

Orbital Rotation Gradient and Hessian for a General JK-only Functional

Setup

Fixed natural occupations n_p , $D_{pq} = n_p \delta_{pq}$. A JK-only two-electron energy functional with a general symmetric occupation-coupling function $f(n_a, n_b) = f(n_b, n_a)$, $f(n, n) = n$, replacing the two-particle density matrix elements as

$$\Gamma_{pqpq} = \frac{1}{2} n_p n_q \quad (\text{Hartree/direct}), \quad \Gamma_{pqqp} = -\frac{1}{2} f(n_p, n_q) \quad (\text{exchange}), \quad (1)$$

so that

$$E_2 = \frac{1}{2} \sum_{pq} n_p n_q \langle pq|pq \rangle - \frac{1}{2} \sum_{pq} f(n_p, n_q) \langle pq|qp \rangle. \quad (2)$$

The Müller functional is the special case $f(n_a, n_b) = \sqrt{n_a n_b}$.

Important: this Γ does *not* satisfy the fermionic antisymmetry $\Gamma_{pqrs} = -\Gamma_{qprs}$ of a genuine two-particle density matrix (it is an N -representability-violating reconstruction, valid only in the idempotent/integer-occupation limit). The gradient and Hessian below are therefore derived directly from the explicit energy expression above (differentiating each integral leg under orbital rotation), *not* from the operator-commutator machinery used for the general Γ case, since that machinery relied on antisymmetry at two separate steps.

Gradient

Define

$$V_{qp} \equiv h_{qp} + \sum_t n_t \langle tq|tp \rangle. \quad (3)$$

$$g_{pq} = (n_q - n_p) V_{qp} + \sum_t [f(n_p, n_t) - f(n_q, n_t)] \langle qt|tp \rangle \quad (4)$$

Hessian

$$\begin{aligned} G_{pq,rs} = & (n_q - n_p) \left[-\delta_{qr} V_{sp} + \delta_{ps} V_{qr} + (n_s - n_r) \langle sq|rp \rangle \right] \\ & - \delta_{qr} \sum_t [f(n_p, n_t) - f(n_r, n_t)] \langle st|tp \rangle + \delta_{ps} \sum_t [f(n_s, n_t) - f(n_q, n_t)] \langle qt|tr \rangle \\ & + [f(n_p, n_s) - f(n_p, n_r) - f(n_q, n_s) + f(n_q, n_r)] \langle qs|rp \rangle \end{aligned} \quad (5)$$

Special case (Müller): set $f(n_a, n_b) = \sqrt{n_a n_b}$ everywhere above.

Status note. This derivation does not rely on 2-RDM antisymmetry (appropriately, since the JK-only Γ lacks it), but it has *not* been independently cross-checked by a second derivation route or a symbolic tool. Given the nonlinear occupation-coupling f and the amount of index bookkeeping, a numerical finite-difference validation of $E(\kappa)$ against this gradient and Hessian (small system, generic fractional occupations, both for $f = \sqrt{}$ and at least one other f) is strongly recommended before use in production code.